MIL-B-2213C(ME)
20 April 1973
SUPERSEDING
MIL-B-2213B(MO)
27 April 1966

MILITARY SPECIFICATION

BRIDGE CONVERSION SET, FIXED BRIDGE:

BAILEY; CONSTRUCTION OF ONE PANEL CRIB PIER,

31 FEET 7 INCHES HIGH, 145-TON

1. SCOPE

1.1 This specification covers structural components of Bridge Conversion Set, Fixed Bridge: Bailey; Construction of 1 Panel Crib Pier, 31 Ft. 7 In. H, 145 Ton (Set No. SC 5420-97-CL-E29).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-S-111	- Screw, Wood.
FF-T-791	- Turnbuckle.
FF-W-92	- Washers, Metal, Flat (Plain).

FSC 5420

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QQ-S624	- Steel Bar, Alloy, Hot Rolled and Cold Finished, (General Purpose).
QQ-T-830	- Tube, Steel, (Carbon, Mechanical; Seamless and Welded).
RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
TT-W-535	- Wood Preservative; Fluoride-Chromate, Arsenate-Phenol Mixture.
TT-W-571	- Wood Preservation: Treating Practices.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-R-52243	- Retainer, Bridge Pin.

STANDARDS

Federal

FED. STD. No. 595 - Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

DRAWINGS

ME

D2674-2 and -5	- Bridge, Fixed: Steel, Portable Panel - M2 Widened Roadway (12'-6") (Bailey Type).
D2738-1 through -6	- Portable Panel Bridge (Bailey Type). Parts for Crib Construction.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A27 - Mild- to Medium-Strength Carbon-Steel Castings for General Application.
A36 - Structural Steel.
A108 - Cold-Finished Carbon Steel Bars and Shafting.
A120 - Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A183 - Heat-Treated Carbon Steel Track Bolts and Carbon Steel Nuts.
A235 - Carbon Steel Forgings for General Industrial Use.
A441 - High-Strength Low-Alloy Structural Manganese Vanadium Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTHERN PINE INSPECTION BUREAU

Standard Grading Rules for Southern Pine Lumber.

(Application for copies should be addressed to the Southern Pine Inspection Bureau, Box 1170, New Orleans, LA 70150.)

WESTERN WOOD PRODUCTS ASSOCIATION

Standard Grading Rules.

(Application for copies should be addressed to the Western Wood Products Association, 700 Yeon Building, Portland, OR 97204.)

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure.
D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The structural components shall be as shown on the following drawings, and as specified (see 6.2):

Drawing D2674-2

BRACE, sway
PIN, sway brace

Drawing D2674-5

BOLT, bracing
BOLT, connector chord
CLAMP, transom
FRAME, bracing, bridge
PIN, connector, panel, 8-5/16 in. lg.
PLATE, tie
RAKER, side strut

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Drawings D2738-1 through -6

BEARING, crib
BEARING, junction link
CAPSILL, crib
CHESS, junction, M2
CLAMP, chord
LINK, junction
LINK, launching nose, MK II
POST, junction, span, female
POST, junction, span, male

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a structural component(s), as applicable, for examination within the time frame specified (see 6.2), to prove prior to starting production, that his production methods will produce structural components that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Structural steel. Structural steel shall conform to ASTM A36.

3.3.2 Low-alloy structural steel. Low-alloy steel shall conform to ASTM A441.

3.3.3 Steel castings. Steel castings shall conform to ASTM A27, grade 70-40, full annealed.

3.3.4 Steel bars and rods. Steel bars and rods except for the panel connector pins shall conform to ASTM A108, grade 1020.

3.3.5 Steel forgings. Steel forgings shall conform to ASTM A235, class

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C1, annealed or normalized.

3.3.6 Lumber. The minimum grade of lumber shall be southern pine No. 1 dense or No. 1 longleaf conforming to Standard Grading Rules for Southern Pine Lumber, or lumber shall be Douglas fir dense No. 1 conforming to Standard Grading and Dressing Rules for Lumber. Lumber shall not be painted but shall be pressure-treated after all cutting has been completed in accordance with TT-W-571, using fluor-chrome arsenate phenol mixture conforming to TT-W-535, type I, with minimum net retention of 0.50 lb/cu.ft.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required for the operation.

3.5 Turnbuckles. Turnbuckles shall conform to FF-T-791, type I, form optional, class 2, self colored, equipped with one jam nut as shown for type III, except that the size shall be 1-1/8 by 6 inches.

3.6 Chain. Chain shall conform to RR-C-271, type I, grade C, class 5, style 1; rings, when required, shall be type VI.

3.7 Steel pipe. Steel pipe shall conform to ASTM A120, black, standard weight, except that the ends shall not be threaded and the hydrostatic test will not be required.

3.8 Steel tubing. Steel tubing shall conform to QQ-T-830, MT 1020 steel, seamless, electric resistance welded or fusion welded type, as hot rolled or as welded condition, round shape.

3.9 Wood screws. Wood screws shall conform to FF-S-111, type I, styles 1s and 2s, steel.

3.10 Bolts, nuts, and washers.

3.10.1 Bolts. Bolts except for chord bolts shall conform to FF-B-575, type II, grade 2 steel, UNC-2A.

3.10.2 Nuts. Nuts shall conform to FF-N-836, type II, style 4, grade A steel.

3.10.3 Washers. Washers shall conform to FF-W-92, type A, grade I, class A, cadmium or zinc plated.

3.10.4 Chord bolts. Chord bolts shall be fabricated from steel conforming to ASTM A183.

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3.11 Panel connector pin. Panel connector pins shall be fabricated from steel conforming to QQ-S-624, steel number 4140, quenched-and-tempered and cold-finished to produce a minimum yield strength of 125,000 pounds per square inch (psi), minimum ultimate tensile strength of 140,000 psi, and a minimum elongation of 15 percent in 2 inches.

3.12 Bridge pin retainer. The bridge pin retainer for the connector panel pin shall conform to MIL-R-52243, nominal length 5-9/16 inches.

3.13 Transom clamps. Transom clamps shall be either steel forgings or weldments.

3.14 Identification marking. Each component shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.15 Treatment and painting. All parts of the structural components except as specified herein shall be treated and painted in accordance with MIL-T-704, type A, with a lusterless finish added matching color No. 36231 of FED. STD. NO. 595. Transom clamps, pins, bolts, nuts, washers, and threaded surfaces of components shall not be painted but shall be coated with preservative (see 5.1.1).

3.16 Stenciling. The nomenclature, Federal Stock number, weight, and cubage shall be stenciled on each major structural component on one side and one end of the part. The letter "NET WT" shall precede the numerals in the weight and the letter "CU" shall precede the cubic displacement. Markings shall be gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.17 Workmanship. The fabricated structural component shall be free of sharp edges, slivers, or burrs.

3.17.1 Steel fabrication. Steel used in the fabrication of the structural components shall be free from laminations, kinks, and sharp bends. Steel shall be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. The straightening of material shall be done by methods that will not cause injury to the metal. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.17.2 Welders and welding.

3.17.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the

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contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.17.2.2 Welding. The surface of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1 - Structural Welding Code." All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

3.17.2.2.1 Welding jigs. Components shall be assembled in steel jigs or frames and welded while held in position. The jigs or frames shall be of a design that will limit the distortion to an amount that can be removed without changing the physical characterisitcs of the metal of the parts being welded.

3.17.3 Pin connections. All pin holes shall be accurately drilled or bored full size through steel drilling templates after all welding has been completed. As an alternate the pin holes may be subpunched in the plates or shapes or cored undersize in the castings prior to assembly and welding, and then reamed to full size through steel drilling templates after assembly and welding. The cored or subpunched holes shall have a maximum diameter 3/8 inch less than the hole size specified and shall be accurately located so that the entire surface of the cored or subpunched hole will be cut during reaming. The holes shall be drilled, bored, or reamed straight through the various parts and shall be within the tolerances specified for position and size. Special care shall be taken to keep panel pin holes in line where so noted on the drawings. The steel drilling template shall be equipped with hardened steel drilling bushings accurately located.

3.17.4 Interchangeability. Like components shall be fabricated to

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be duplicates within the tolerances specified and shall be interchangeable.

3.17.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Toelrances for positioning of holes and parts shall be referenced from one end or side of the member and shall not be cumulative.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Structural components which conform to all requirements in section 3 and 5 of this specification and pass all applicable examinations in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance or rejection (see 6.4 and 6.5).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

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4.3 Preproduction inspection.

4.3.1 Examination. The preproduction structural component(s) shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each structural component shall be inspected for conformance to the dimensions covered by the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The structural component shall be examined as specified herein for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor

201. All dimensions not designated major on the SQAP not as specified.
202. Identification marking, incorrect, missing, or illegible.

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203. Treatment and painting not as specified.
204. Stenciling not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Sample selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
107. Unpainted metal surfaces requiring a contact preservative in accordance with the referenced document not coated with the specified preservative.
108. Components of unlike description packed together for level A or B.
109. Gross weight exceeds the weight limitation of the box for level A or B.
110. Contents not blocked, braced, anchored, or immobilized in accordance with the referenced document for level A or B.
111. Strapping not zinc coated for level A.
112. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation (see 6.6).

5.1.1 Unprotected surfaces. Unpainted metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The components of like description, preserved as specified in 5.1, shall be packed together in close-fitting boxes

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conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional. The quantity of components of like description to be packed together shall be determined by the gross weight limitation of the box, in accordance with the applicable box specification. The components shall be blocked, braced, anchored, or otherwise immobilized within the box in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated, and on boxes provided with skids, placed adjacent to the skids.

5.2.2 Level B. The components of like description, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except boxes shall be domestic type or class as applicable and strapping is not required to be zinc coated.

5.2.3 Level C. The components shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in wooden containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. (see 6.6) Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The structural components are used in conjunction with standard bailey bridge parts to make up a panel crib pier and to provide for a flexing action at supports.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Component required (see 3.1).
(c) Time frame required for submission of the preproduction model (see 3.2).
(d) Level of packing required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production structural components from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish structural components conforming to this specification.

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6.4 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary Quality Assuarnce Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Preservation. Levels of preservation have not been included as the requirements specified are the minimum acceptable for any level of protection. For purposes of preservation/packing level marking, the marking shall be A/A, A/B, or A/C as applicable.

Custodians:

Army - ME

Preparing activity:

Army - ME

Project No. 5420-A234

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-2213C(ME) Bridge Conversion Set, Fixed Bridge: Bailey; Construction of one Panel Crib Pier, 31 Feet 7 Inches High, 145-Ton

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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